SUPPLEMENTARY MATERIAL: COMPETITION BETWEEN RADIAL DIFFUSION AND INTERFACIAL EXCHANGE IN THERMAL-GRADIENT DEPOSITION COLUMNS

SUPPLEMENTARY MATERIAL TO THE WORKING MANUSCRIPT

Jan Kren[1] **and Bojan Ničeno**[1]
[1]Paul Scherrer Institute, 5232 Villigen PSI, Switzerland

14 October 2026

This document collects the case inventory, the detailed material-ledger and solver diagnostics, the scalar mesh checks of the original upwind campaign, the comparison-bin-width analysis, the property and thermodynamic records and the complete Graetz–Robin verification table. Equations, tables and sections of the main text are referred to by name. All numbers are generated from the same saved results as the main text (`manuscript/prepare_results.py`, `manuscript/assess_resolution.py`).

S1 Case inventory

Table S1 lists the 64 follow-up cases of the original 18,432-cell upwind campaign. They cover the following cases:

- mesh and time-step checks: pure PbI_2 and three steel Q2 cases (baseline thermodynamics at 0.1% and 1% metal saturation, and shifted BiI thermodynamics at 1%);
- physical transport, accommodation, source-position and flow-reference variations: pure PbI_2, the two steel Q2 cases with baseline thermodynamics, and silica Q2 at 0.1%;
- extra source-loading cases: both columns and both thermodynamic branches;
- temperature-coordinate variants: silica Q1 and silica Q2 at 0.1%.

This is a structured set of conditional comparisons, not a factorial uncertainty study. Each numerical comparison changes one resolution setting at a time.

Table S1: Completed follow-up matrix: number of cases and the variation applied in each study. The 15 baseline cases serve as additional references. Flow-stop cases continue a 60 s baseline to 80 s.

Study	Cases	Variation
Mesh and time step	12	Three grids; 2 and 1 ms
Gas diffusion	12	$D/2$, $2D$, PbI_2–He correlation
Iodide accommodation	8	$\alpha = 0.1$, 0.01
Source position	8	$\Delta x_s = \pm 1.5$ cm
Q2 metal loading	12	$\beta = 0.002$, 0.003162, 0.005623
Flow reference	4	273.15 versus 298.15 K
Silica temperature coordinates	4	$\Delta x_T = -4.10$, -4.65 cm
Flow-stop relaxation	4	60–80 s; fixed temperature

The source-position alternatives move the release region by $\pm 1.5\,\mathrm{cm}$ without adding a boat obstruction. The flow-stop cases restart from the saved 60 s fields, remove injection, set the carrier flow to zero and impose zero normal species flux at both ends. The original nonuniform temperature profile stays fixed. These cases therefore measure how residual gas redistributes during a 20 s flow stop at fixed temperature; they are neither a clean-helium purge nor a full cooling calculation.

Figure S1 shows the largest baseline-relative iodine profile change in each physical-input category of the main-text sensitivity table.

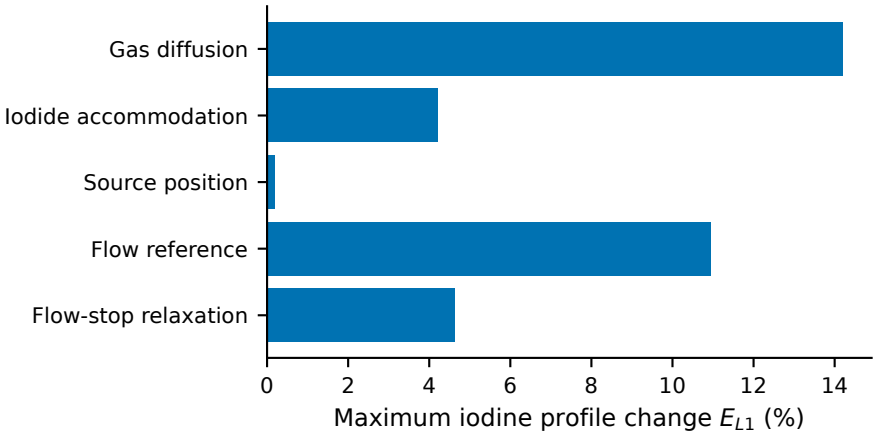


Figure S1: Maximum baseline-relative iodine profile change E_{L1} in each selected physical-input category (upwind follow-up cases on 18,432 cells). The flow-stop value uses an 80 s endpoint; the others use 60 s. These maxima describe the tested case set, not a general uncertainty budget or an experimentally identified parameter ranking.

Six further sets of runs complete the evidence, all with the same solver executable. (i) The 19 production cases repeat the 15 baselines and the four silica temperature-coordinate cases on 41,472 cells with the limited-linear scheme; all other inputs are unchanged. (ii) A scheme and mesh study takes two cases, pure PbI_2 and the shifted-BiI steel case at 1% metal saturation, which had the largest upwind fine/medium change. It runs them with upwind, limited-linear and van Leer advection on 8,192 to 41,472 cells, and the steel case with upwind and limited-linear also on 93,312 cells. This adds 12 runs to the existing upwind levels and the production runs. Pure PbI_2 and the two unshifted steel Q2 cases were also run with upwind on 93,312 cells, and a repeat of the shifted-BiI upwind run on that mesh reproduced it bitwise. (iii) Four original 18,432-cell upwind cases, the three baselines and the shifted-BiI steel case at 1% metal saturation, are repeated over the full 60 s with eight instead of three species correctors. (iv) Ten runs simulate five single-species PbI_2 experiments of Lobresco et al. [1] on two meshes. (v) The diffusivity ($D/2$, $2D$) and accommodation ($\alpha = 0.1$, 0.01) variations of pure PbI_2 and steel Q2 at 1% metal saturation are repeated with the production scheme on 41,472 cells (8 runs). (vi) 13 runs of a plane channel vary the accommodation coefficient over six decades for the Damköhler sweep of the main text, and 5 repeat five of them with twice the wall-normal resolution. The stand-alone BiI_3 experiment at 800 °C was not run because its axial temperature profile is not published.

S2 Material ledger and solver diagnostics

The frozen carrier keeps the converged face mass flux of the `rhoSimpleFoam` solution rather than a flux reconstructed from interpolated velocity. At these low mass flows, a zero-initialised potential correction removes residual discrete divergence before species transport. Carrier preparation rejects a carrier if its original cell or global flux defect exceeds 10^{-6} of the inlet flow, or if a flux or velocity correction exceeds 10^{-5} of the corresponding scale. After correction, inlet, global and cell continuity and leakage through impermeable boundaries are checked against 10^{-12} of the inlet flow. This procedure is neither a new physical flow model nor evidence that the velocity field has converged.

Element conservation is examined in two ways: through the solver ledger and through the independently formed raw material residue of the main text (Eq. “balance”). The ledger also records the linear-solve defect and any clamping correction separately. These terms explain bookkeeping closure, but they are not inserted into the residue when the raw material balance is assessed.

All residues are divided by the supplied material and taken as the largest value over recorded steps and represented elements. The largest raw residue is 1.41×10^{-12} for the 19 production cases (41,472 cells, limited-linear advection) and 8.85×10^{-13} for the 60 source-driven follow-up cases; the 18,432-cell upwind baselines had 5.97×10^{-13}. For the follow-up cases, the largest accounted closure is 9.60×10^{-16} and the largest recorded solver defect is 8.64×10^{-13}; for the production cases they are 1.12×10^{-15} and 1.45×10^{-12}. The four flow-stop cases keep their restart ledger and are excluded from this summary of newly accumulated source material.

Canonical initialisation of the equilibrium solver makes a restarted case independent of an unsaved equilibrium warm state. The baseline logs report no gas-equilibrium solver failures and a maximum equilibrium element residue of 6.43×10^{-16}. Input-validity and solver-success counters are distinct. At startup, sign flags caused equilibrium to be skipped in up to 1,170 cells on one step, and the maximum negative iodine diagnostic was 1.14×10^{-124} of supplied iodine. Matched five-step startup replays identified extremely small leading-tail inventories at these flags; no flagged cells remained at the final step. The calculations should therefore not be described as free of negative trace values, even though the reported amount is immaterial to the macroscopic balance. The sum of the individual species mole-fraction maxima was at most 9.19×10^{-5}. Because these maxima need not coincide, this is an upper bound on the reactive trace loading, not a measured mixture mole fraction at one location.

Some paired wall-coupling iterations did not meet the requested 10^{-16} outer criterion. This is reported separately from equilibrium failure, the final transport defect and the inventory-bound checks. To test its effect, matched restarts of four representative cases at 59 s compared three and eight correctors, and outer tolerances of 10^{-16} and 10^{-14}, over 20 time steps (40 ms). The largest species-field change, scaled by its control maximum, was below 3.6×10^{-8}, and the changes in new wall increments were negligible. In the four full-run checks with eight correctors the largest outer initial residual is still 2.4×10^{-15}–4.4×10^{-15}, and nearly every step ends at the corrector limit, as with three: the requested 10^{-16} lies below the attainable residual floor. No manufactured-solution test of the full coupled system is presented; the Graetz–Robin test of the main text verifies the wall discretisation only.

S3 Scalar mesh checks and binned maxima

Three-grid scalar checks follow the distinction between refinement trends and an established asymptotic range [2–4]. For the upwind follow-up triplets (8,192, 18,432 and 41,472 cells), 17 of the 34 nontrivial or zero-valued combinations of species, total, centroid and width are oscillatory, four have nondecreasing differences, one is unresolved at round-off and 12 admit only conditional Richardson/GCI estimates. With upwind advection the shifted-BiI steel case changes by essentially the same amount on the first two refinements (10.47% and 10.58%). On 41,472 cells, van Leer and limited-linear advection differ by 1.16% (steel) and 0.27% (pure PbI_2).

With the production scheme (41,472 cells, limited-linear) the pure PbI_2 binned maximum stays at 36 cm and 366.4–366.5 °C on all three levels, and on 41,472 cells the 37 cm bin is 23% lower; yet the maximum of the 3 cm smoothed profile is at 37 cm and 350.5 °C, and the wall temperature at the deposit centroid (37.11 cm) is 348.7 °C. In the upwind steel Q2 0.1% case, the dominant BiI_3 bin moves from 57 to 58 cm, and its reported temperature changes from about 166.4 to 138.8 °C: a one-bin switch changes the peak temperature by nearly 28 K without a comparable displacement of the profile as a whole.

The local one-dimensional estimate $D_{\mathrm{num}} \simeq |u_x|\Delta x/2$ of the upwind numerical diffusion is a Taylor-series screening estimate, not a measured truncation error for the radial, variable-density coupled problem; it exceeds the prescribed $10^{-4}\,\mathrm{m^2\,s^{-1}}$ where $|u_x|$ exceeds 0.061 m/s on the 18,432-cell mesh.

S4 Dependence on comparison-bin width

To distinguish sensitivity of the reported observable from a change in the CFD mesh, the saved 1 cm iodine inventories of the original upwind campaign were summed into groups of two and four consecutive bins. Every possible integer-bin offset was tested for each width, retaining partial end groups and the full deposited inventory. No CFD field was recomputed and no sub-bin reconstruction was introduced. Table S2 gives the fine/medium inventory differences for these coarser observables. Ranges reflect bin origin, not statistical uncertainty.

Table S2: Original upwind fine/medium iodine inventory-profile differences (percent, 41,472 versus 18,432 cells) after aggregation of native bins. The 2 and 4 cm ranges cover all respective integer-bin offsets. W_1 uses the original 1 cm bin centres and separately normalised profiles; it measures redistribution distance, not position error against the experiment.

Case	1 cm	2 cm	4 cm	W_1 (cm)
Pure PbI2	5.06	4.09–4.81	1.22–4.81	0.0610
Steel Q2, 0.1%	5.26	1.77–4.12	0.51–2.45	0.0311
Steel Q2, 1%	2.06	0.85–1.43	0.11–1.31	0.0115
Steel Q2, shifted BiI	10.58	3.37–10.02	0.52–10.02	0.0711

For the shifted-BiI case, a 4 cm comparison gives differences from 0.52% to 10.02% depending on the grouping origin, so a favourable aggregation can hide a substantial local difference. Summing signed bin differences before taking their absolute values necessarily allows cancellation; a reduced aggregated L1 is not additional convergence evidence.

As a complementary position-sensitive measure, let p_b and q_b be the fine and medium iodine inventories normalised separately over the whole column, and x_b their common native bin centres. The one-dimensional transport distance for these discrete distributions is

$$W_1 = \sum_{b=1}^{B-1} \left| \sum_{j=1}^{b} (p_j - q_j) \right| (x_{b+1} - x_b). \tag{S1}$$

It is the minimum mean distance needed to redistribute one normalised binned inventory into the other. Values of 0.0115–0.0711 cm indicate small average spatial redistribution between the two tested meshes, despite larger local L1 differences. This is a distance between computed binned profiles, not a bound on continuum error or evidence that the experimental position uncertainty has been resolved. Four analytical tests (edge-inventory preservation, signed-change cancellation on aggregation, amplitude invariance of the shape measures and a known point-mass displacement) check this post-processing, not the CFD solver.

S5 Properties, thermodynamic provenance and units

This section lists the helium properties, molar masses, thermodynamic records and optional PbI_2–He diffusivity used in the calculations.

S5.1 Helium and transported molecular masses

The executed helium carrier uses $M_{\mathrm{He}} = 0.0040026\,\mathrm{kg\,mol^{-1}}$, $C_p = 5193\,\mathrm{J\,kg^{-1}\,K^{-1}}$ and

$$\rho = \frac{pM_{\mathrm{He}}}{RT}, \qquad \mu(T) = \frac{A_s\sqrt{T}}{1 + T_s/T}, \tag{S2}$$
$$A_s = 1.4683 \times 10^{-6}, \qquad T_s = 83.40\,\mathrm{K}.$$

The numerical value of A_s gives viscosity in Pa s when T is in K. These are case inputs to the perfect-gas, constant-heat-capacity, Sutherland model, not helium properties newly fitted to measurements. Partial-pressure conversions use absolute pressure, while thermodynamic activities use $p^\circ = 10^5\,\mathrm{Pa}$. The assumed controller reference pressure, 101,325 Pa, is a separate quantity. Table S3 lists the species masses used in transport and stoichiometric conversion.

Table S3: Molar masses of the nine transported gases. Condensed iodide inventories use the same formula masses; the shared Pb and Bi inventories use atomic masses.

Gas	kg/mol	Gas	kg/mol	Gas	kg/mol
Pb	0.207200	Bi	0.2089804	I	0.12690447
PbI	0.33410447	Bi_2	0.4179608	I_2	0.25380894
PbI_2	0.46100894	BiI	0.33588487	BiI_3	0.58969381

S5.2 Thermodynamic records and phase treatment

Pb–I–He functions are evaluated from external NASA nine-coefficient records labelled with Gurvich source information. Bi–I functions use an external ThermoFun compilation labelled HERACLES/Barin. A common property evaluator converts all records to a consistent absolute convention, $G^\circ(T) = H^\circ(T) - TS^\circ(T)$, before forming reaction Gibbs energies. Mixing this convention with an apparent formation Gibbs energy would give inconsistent equilibrium constants. The solver tables store formation constants and phase-resolved vapour pressures as logarithms. Input-file hashes, record identifiers and table-generation scripts are recorded in the benchmark provenance.

For an iodide with one gas molecule per condensed formula unit,

$$p_{\mathrm{eq}}(T) = p^\circ \exp\left[-\frac{G_g^\circ(T) - G_c^\circ(T)}{RT}\right], \tag{S3}$$

evaluated on the stable available condensed branch. The BiI_3 liquid branch is constructed at $T_{\mathrm{fus}} = 680.85\,\mathrm{K}$ from the crystalline record with $\Delta H_{\mathrm{fus}} = 31.798\,\mathrm{kJ\,mol^{-1}}$ and $C_{p,l} = 150.624\,\mathrm{J\,mol^{-1}\,K^{-1}}$, as recorded in the input-generation metadata. It is a constructed phase record, not an independently confirmed liquid thermodynamic dataset. Bi and Bi_2 share condensed Bi but have different gas chemical potentials and stoichiometry.

The BiI +29 kJ/mol variant modifies the gas record and regenerates all formation and disproportionation constants consistently. Both the baseline and the perturbed functions are hypotheses for the sensitivity calculation; the BiI source record still needs independent confirmation. The external coefficient files are not reproduced in the manuscript data package. Numerical conservation of the model derived from them cannot establish their physical correctness or complete the provenance assessment.

S5.3 Optional PbI_2–He diffusion estimate

Only the PbI_2–He transport variant uses the following implemented correlation:

$$D_{\mathrm{PbI_2-He}} = \frac{1.86\times10^{-7}T^{3/2}\sqrt{1/4.0026+1/461}}{(p/101325)\sigma^2\Omega(T^*)}, \tag{S4}$$

$$T^* = T/71.117857, \qquad \sigma = 4.258468, \tag{S5}$$

$$\Omega(T^*) = 0.5313 - 1.2946e^{-0.9922T^*} + \frac{1.5591}{T^*} - \frac{0.1918}{(T^*)^2}. \tag{S6}$$

In this numerical expression T is in K, p in Pa, σ in Å and the masses in g/mol; the resulting D is in $\mathrm{m^2\,s^{-1}}$. To reproduce the original correlation, the implementation keeps 461 g/mol rather than the more precise transport formula mass in table S3. Its estimated collision parameters follow the original polarisability/electron-count construction recorded in the code. The other eight gases keep the constant baseline coefficient in this variant. The correlation is a screening alternative, not an experimentally validated binary diffusivity or a replacement for all missing Pb–Bi–I transport data.

S6 Lobresco et al. comparison: set-up details

The five cases of Lobresco et al. [1] are computed on a 0.69 m, 5° wedge of 4.8 mm inner diameter whose origin is taken to be the figure origin of that study (clean helium enters at $x = 0$ at the wall temperature there). The production mesh has 22,896 cells (318×72), the axial and radial spacing of the 41,472-cell Liu mesh, and the coarse mesh 10,176 cells (212×48), that of the 18,432-cell mesh. The time step is 2 ms, 0.5 ms at 540 mL/min. The sample is an evaporating inventory of the Table 1 amount (times $5/360$ for the wedge), a cylinder 1 cm long and 0.48 mm in radius on the axis at the digitised sample arrow (0.13 m at 106 mL/min, 0.15 m at 5 and 540 mL/min); this geometry is an assumption. The 106 mL/min wall temperature is the mean of the curves of the T106_6 and T106_300 panels. Helium properties are those of the Liu cases, not the NIST tables of that study. Carriers are audited to 10^{-12} of the inflow; the largest absolute ledger closure over all steps and runs is 3.3×10^{-16} of the inventory.

A wall-temperature table distributed with the T-Flows test case of that study differs from the published curves: its maximum of 1210 K lies at 0.04 m, whereas the figures peak at 1060–1070 K at 0.13–0.15 m, and between 0.30 and 0.60 m it differs from them by up to 68–92 K. We use the published curves.

S7 Graetz–Robin verification: all runs

Table S4 lists every three-level sequence of the Graetz–Robin verification (Pe = 5, 45 runs including one time-accurate control): Gauss linear advection with zero-gradient and fixed-value outlets, and SuperBee advection, for both wall laws and both Biot numbers. "halfCell" is the half-cell series law of the main text and "none" the kinetic law applied to the cell-centre value. Every fitted decay rate equals that of the developed discrete mode of the documented scheme within 7.09×10^{-9}. Moving either or both ends of the fit window by $\pm1.5R$, or fitting the mixing-cup concentration or the wall uptake instead of the line density, changes μ by less than 10^{-7}; the fixed-value outlet changes it by less than 10^{-8}. The reference error bound is 1.69×10^{-12}. On the coarser 32/64/128 sequence the cell-centre law at Bi = 50 has $\mathrm{GCI_{fine}} = 0.678\%$; this triplet is too coarse for a first-order wall law. The time-accurate control run ($N_r = 32$, Bi = 50, half-cell law, Courant number 1) reaches the decay rate of the discrete mode of its mesh within 10^{-9}. The records (reference, per-run fits, criteria log and provenance, including the solver SHA-256) are in `manuscript/data/graetz` with instructions to regenerate them.

Table S4: Observed order p, relative error of the Richardson-extrapolated decay rate and $\mathrm{GCI}_{\mathrm{fine}}$ [4] (refinement ratio 2, safety factor 1.25) for every three-level sequence of the Graetz–Robin verification.

Scheme	Outlet	Bi	Wall law	N_r	p	$\mu_{\mathrm{ext}}/\mu_{\mathrm{ref}} - 1$	GCI (%)
linear	zeroGradient	1	halfCell	32/64/128	2.0033	1.40×10^{-8}	7.55×10^{-4}
linear	zeroGradient	1	halfCell	64/128/256	2.0010	1.07×10^{-9}	1.89×10^{-4}
linear	zeroGradient	1	none	32/64/128	1.0154	3.50×10^{-5}	3.08×10^{-1}
linear	zeroGradient	1	none	64/128/256	1.0077	8.84×10^{-6}	1.55×10^{-1}
linear	zeroGradient	50	halfCell	32/64/128	1.9718	6.97×10^{-8}	4.06×10^{-4}
linear	zeroGradient	50	halfCell	64/128/256	1.9919	9.47×10^{-9}	1.00×10^{-4}
linear	zeroGradient	50	none	32/64/128	1.0081	3.88×10^{-5}	6.78×10^{-1}
linear	zeroGradient	50	none	64/128/256	1.0045	1.10×10^{-5}	3.41×10^{-1}
linear	fixedValue	1	halfCell	32/64/128	2.0033	1.40×10^{-8}	7.55×10^{-4}
linear	fixedValue	1	halfCell	64/128/256	2.0010	1.09×10^{-9}	1.89×10^{-4}
linear	fixedValue	1	none	32/64/128	1.0154	3.50×10^{-5}	3.08×10^{-1}
linear	fixedValue	1	none	64/128/256	1.0077	8.84×10^{-6}	1.55×10^{-1}
linear	fixedValue	50	halfCell	32/64/128	1.9718	6.97×10^{-8}	4.06×10^{-4}
linear	fixedValue	50	halfCell	64/128/256	1.9919	9.63×10^{-9}	1.00×10^{-4}
linear	fixedValue	50	none	32/64/128	1.0081	3.88×10^{-5}	6.78×10^{-1}
linear	fixedValue	50	none	64/128/256	1.0045	1.10×10^{-5}	3.41×10^{-1}
SuperBee	zeroGradient	1	halfCell	64/128/256	2.0024	4.68×10^{-9}	4.23×10^{-4}
SuperBee	zeroGradient	1	none	64/128/256	1.0143	1.63×10^{-5}	1.54×10^{-1}
SuperBee	zeroGradient	50	halfCell	64/128/256	2.0104	6.44×10^{-8}	1.30×10^{-3}
SuperBee	zeroGradient	50	none	64/128/256	1.0231	5.77×10^{-5}	3.36×10^{-1}

References

[1] Francesco Lobresco, Georg Tiebel, Gian Carmine Conigliaro, Patrick Steinegger, Olha Marinich, Nikolaos I. Prasianakis, Jörg Neuhausen, and Bojan Ničeno. Assessment of temperature-based and vapor-pressure-driven models for CFD simulation of volatile species deposition. *Nuclear Engineering and Design*, 459:115201, December 2026. ISSN 0029-5493. doi: 10.1016/j.nucengdes.2026.115201. URL https://doi.org/10.1016/j.nucengdes.2026.115201.

[2] NPARC Alliance. Examining spatial (grid) convergence. NASA Glenn Research Center CFD verification and validation tutorial, n.d. URL https://www.grc.nasa.gov/www/wind/valid/tutorial/spatconv.html. Accessed 9 October 2026.

[3] P. J. Roache. Perspective: A method for uniform reporting of grid refinement studies. *Journal of Fluids Engineering*, 116(3): 405–413, September 1994. ISSN 1528-901X. doi: 10.1115/1.2910291. URL https://doi.org/10.1115/1.2910291.

[4] Ishmail B. Celik, Urmila Ghia, Patrick J. Roache, Christopher J. Freitas, Hugh Coleman, and Peter E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7): 078001, 2008. ISSN 0098-2202. doi: 10.1115/1.2960953. URL https://doi.org/10.1115/1.2960953.